

Indian Institute Of Technology Gandhinagar



Assignment Report

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Branch: Integrated Circuit Design And Technology (ICDT)

Department: Electrical Engineering

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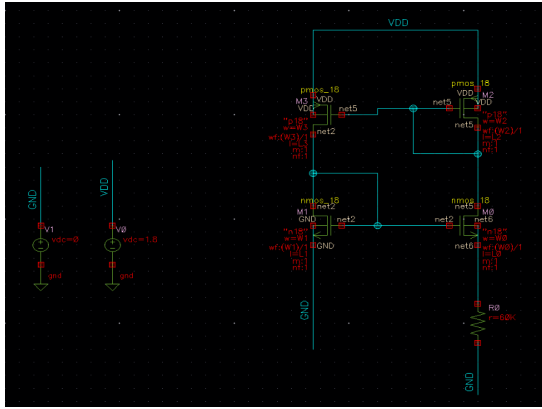
Submitted to: Madhav Pathak

Assignment No: 03

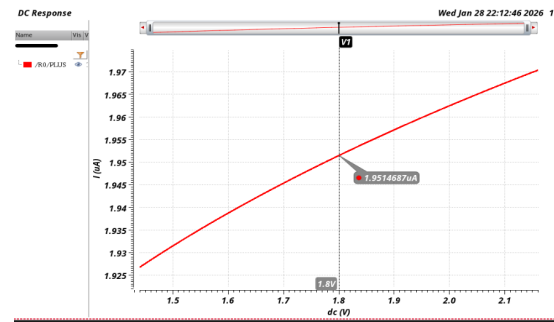
Date: january 25, 2026

$$R = 1.0086 \times 10^6 \Omega$$

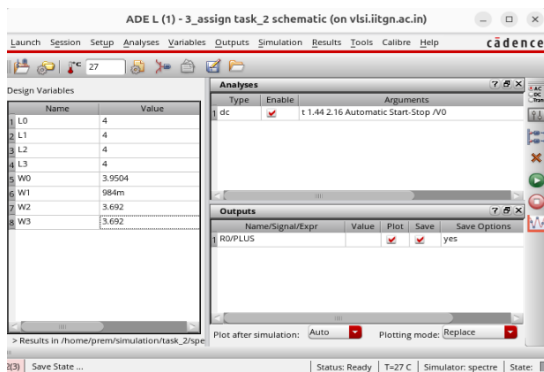
$$R \approx 1.01 \text{ M}\Omega$$



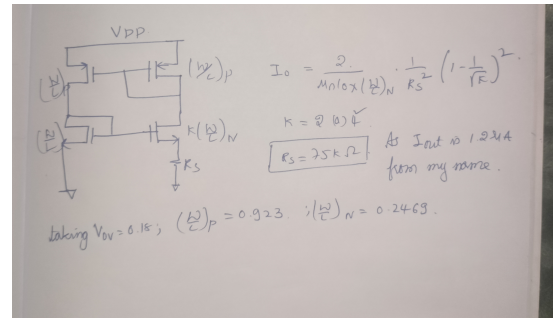
(a) Circuit Diagram



(b) Graph for Iout Vs Vdd



(a) width and length of transistors scaled making ratio constant



(b) Calculations

TASK 2

- Designing Sub-1-V BGR using OTA [3pts]

Using the Temperature Coefficient of CTAT and PTAT you found after simulation in Assignment-2, now design the Sub-1-V BGR (refer to Fig.3) (having the Output BGR Voltage less than 1V) as per the following specifications and report/plot the simulated measurements. (The Output Voltage of the BGR should be according to the table below.)

- Temperature regulation: 10mV (for temperature variation -40 to 125 °C)
- Supply voltage regulation: 10 mV (for supply voltage variation: $\pm 0.1\text{V}$)
- Power budget: 50 μW

- Supply voltage: 1.8V
- PSRR at 1 MHz > 25dB
- Include derivation (digitally written/scanned) in your report and justify your choice of design parameters like value of resistors, multiplier, etc.
- Design your circuit for the tt (typical-typical) process corner. Simulate and analyze your circuit for the four other corners (ff, ss, fs, and sf) and include temperature regulation and supply voltage regulation for all corners in a tabular format in your report. (Necessary to include Corner simulation, but it's fine if specs don't match for different corners)

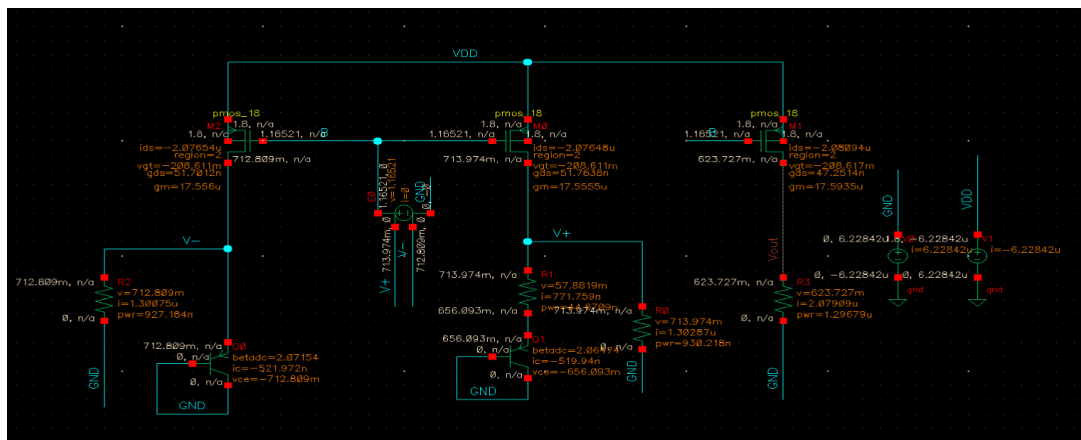


Figure 4: Circuit Diagram for BGR

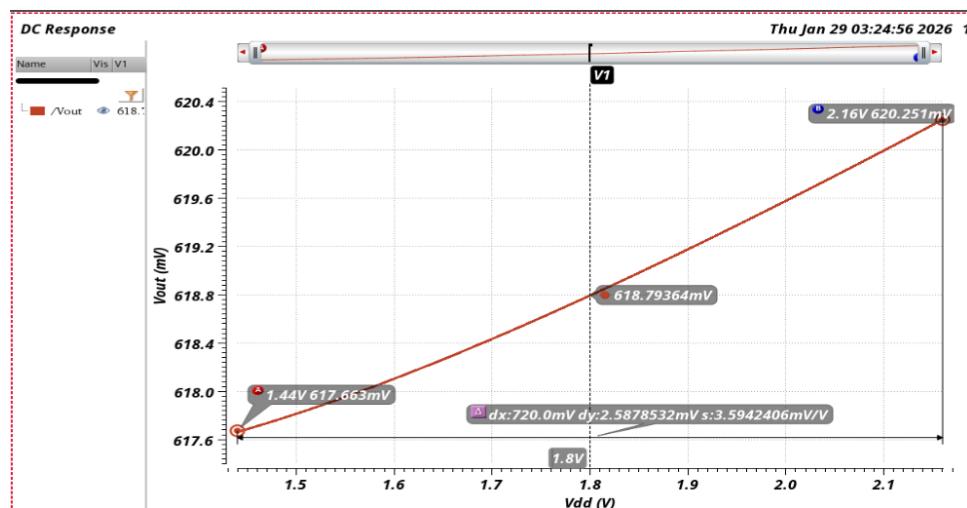


Figure 5: Supply regulation

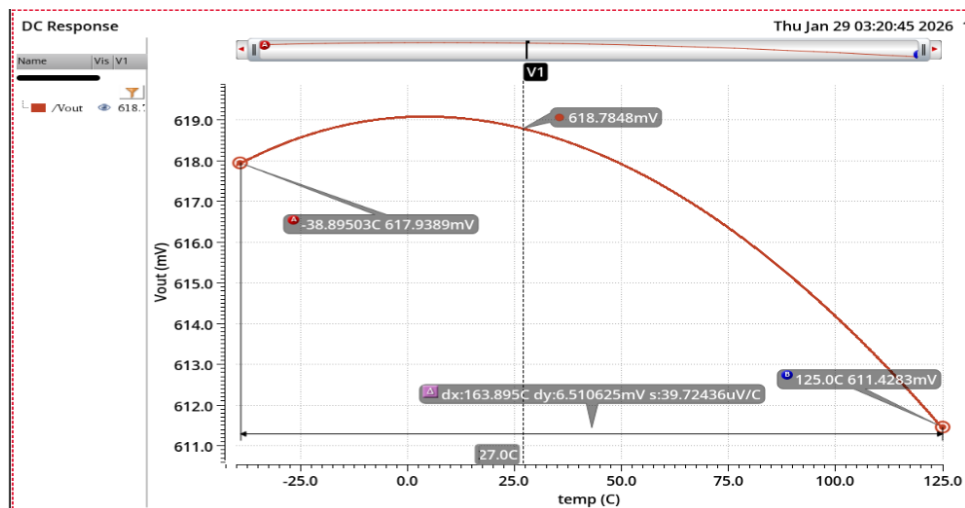


Figure 6: Temperature regulation

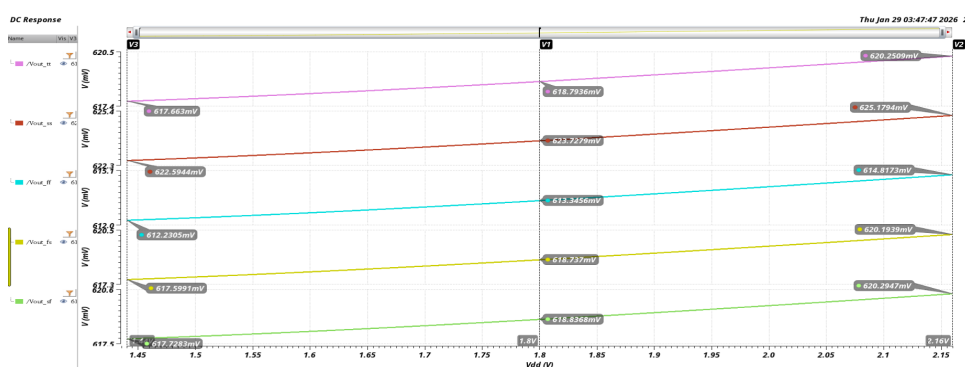


Figure 7: Process corners for supply regulation

Table 1: DC Supply Sweep Results Across Process Corners

Corner	V_{out} @ 1.45 V (mV)	V_{out} @ 1.8 V (mV)	V_{out} @ 2.15 V (mV)	Max-Min (mV)
TT	617.663	618.7936	620.2509	2.5879
SS	622.5944	623.7279	625.1794	2.5850
FF	612.2305	613.3456	614.8173	2.5868
FS	617.5991	618.7370	620.1939	2.5948
SF	617.7283	618.8368	620.2947	2.5664

Across all five process corners (TT, SS, FF, FS, SF), the output voltage variation over the supply range 1.45 V to 2.15 V is approximately 2.56–2.60 mV. The worst-case supply-induced variation is 2.595 mV (FS corner), which is well within the 10 mV supply regulation specification

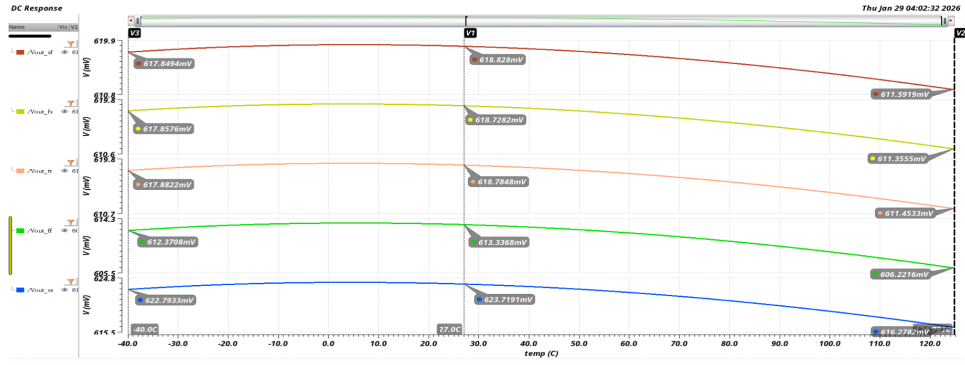


Figure 8: Process corners for temperature regulation

Table 2: Temperature Sweep Results Across Process Corners

Corner	V_{out} @ -40°C (mV)	V_{out} @ 27°C (mV)	V_{out} @ 120°C (mV)	Max-Min (mV)
SF	617.8494	618.8282	611.5919	7.2363
FS	617.8576	618.7282	611.3555	7.3727
TT	617.8822	618.7848	611.4533	7.3315
FF	612.3708	613.3368	606.2216	7.1152
SS	622.7933	623.7191	616.2782	7.4409

For the temperature range -40°C to 120°C , the output voltage variation across process corners lies between 7.12 mV and 7.44 mV. The worst-case temperature-induced variation occurs in the SS corner (7.44 mV), which remains below the 10 mV temperature regulation requirement.

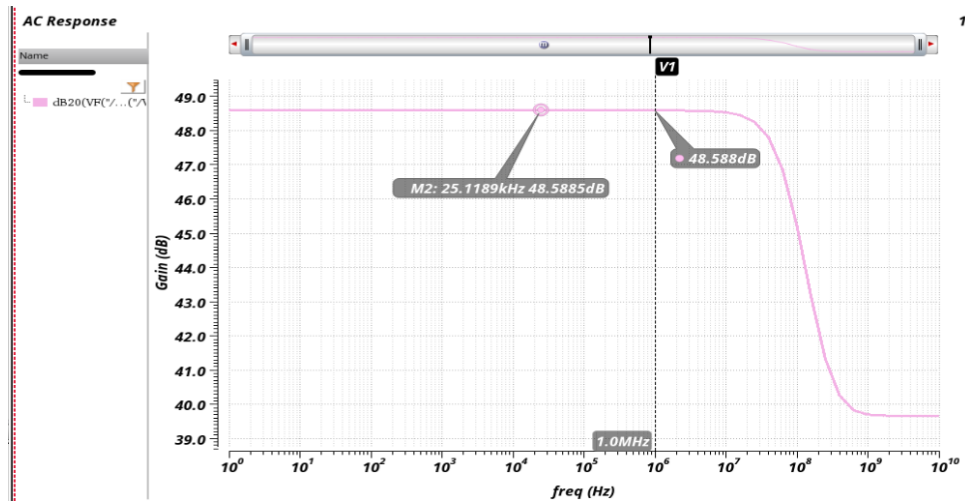


Figure 9: PSRR

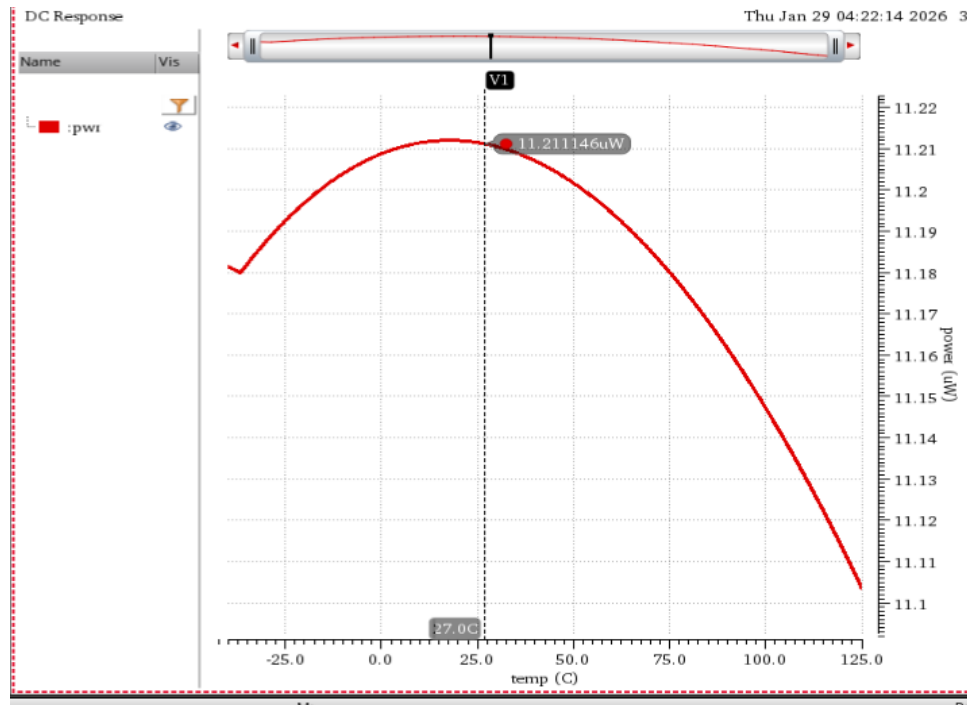


Figure 10: power with respect to temperature

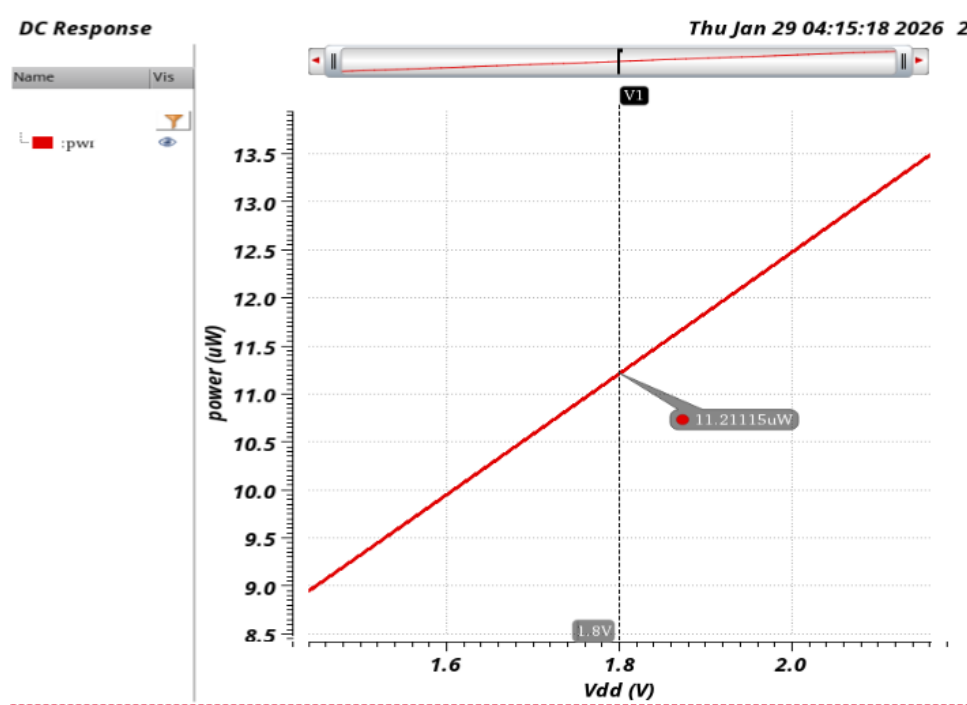


Figure 11: power with respect to supply

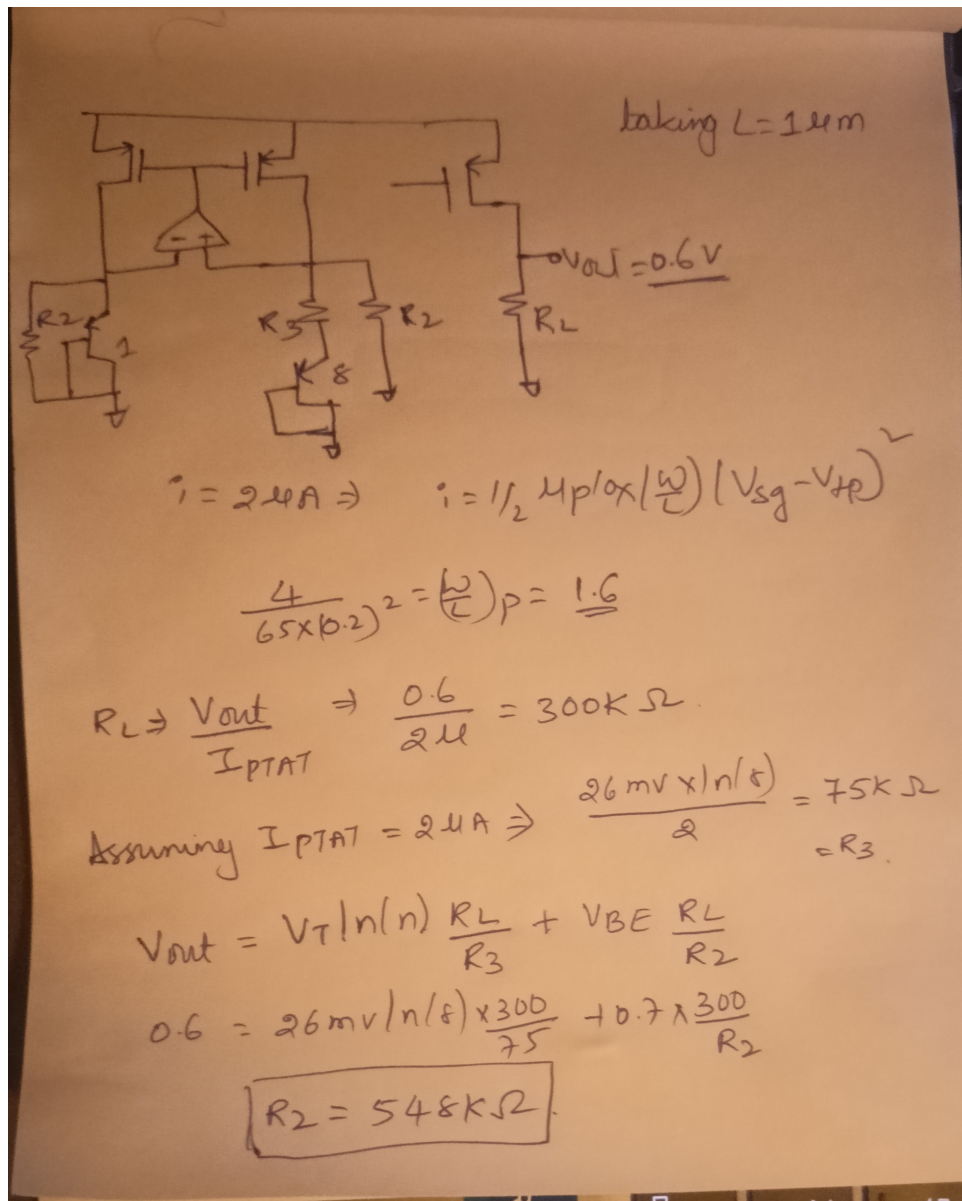


Figure 12: Calculations

Resistor	Calculated values	$\frac{w}{l}$	Vov(mV)
R_1	300k Ω	1.6	208.611
R_2	548k Ω	1.6	208.611
R_3	75k Ω	1.6	208.611

Table 3: Calculated values of Resistors

Table 4: Summary of Bandgap Reference Performance

Parameter	Value	Remarks
Output Voltage (V_{out})	0.6 V	Sub-1-V operation
PSRR @ 1 MHz	48.5 dB	Exceeds specification
Supply Regulation	2.58 mV	VDD sweep
Temperature Regulation	6.51 mV	-40°C to 120°C
Supply Voltage	1.8 V	Nominal

Conclusions

The designed bandgap reference demonstrates robust overall performance across DC, AC, supply, and temperature analyses. The output voltage is centered around 0.6 V, exhibiting a low supply-voltage sensitivity of 2.58 mV over the applied VDD sweep, and a temperature variation of 6.51 mV from -40°C to 125°C , confirming effective CTAT–PTAT compensation. The PSRR reaches 48.5 dB at 1 MHz, significantly exceeding the required specification, indicating strong immunity to supply noise in the operating frequency range. Power consumption remains low, with a peak value of approximately 11.2 μW around room temperature and decreasing at higher temperatures, well within the allocated power budget. Overall, the results verify that the proposed BGR meets all target specifications for output accuracy, regulation, noise immunity, and power efficiency across operating conditions.